

Miriam Elia, *PhD*

Designing the systems behind trustworthy AI.

Email ↗ Website ↗ LinkedIn ↗ GitHub ↗ Google Scholar ↗



Profile

Responsible AI researcher specializing in high-risk AI lifecycle design. I design, evaluate, and structure AI systems across technical, regulatory, and organizational boundaries, and translate that work for interdisciplinary and public audiences. Over four years of doctoral work, I built education programs, ran cross-institutional projects, and engaged with AI governance ecosystems across Europe and Southeast Asia. Based in Munich, open to relocation.

Career Path

- 2021 – 2026** **Doctor of Natural Sciences (PhD), University of Augsburg.** Chair of Software Methodologies for Distributed Systems. Defended *summa cum laude*, October 2025.
- › Designed *MQG4AI*, a Quality Gates methodology for Responsible AI lifecycle design grounded in EU AI Act quality management system requirements (Article 17), illustrated through certifiable AI use cases in medicine.
 - › Conceptualized and led *EsophagusVisualization*, an open-source clinical tool for interactive 3D-reconstructions from multi-modal esophageal imaging; coordinated an interdisciplinary team of computer science students, clinicians, and ethicists.
 - › Drove the *ECG-AI in Emergency Medicine* work with corpuls and the German Heart Center of TU Munich, with a focus on multi-label classification evaluation.
 - › Supervised five BSc and MSc theses and translated research to clinical, legal, and public audiences across Europe.
- 2019 – 2021** **M.Sc. Business Information Systems Engineering, University of Augsburg.** Research assistant at the Institute for Software & Systems Engineering: open-source *Decision Tree Learning* web application with international academic adoption (PMLR conference publication, 2020), and contributions to the Kognia research project.
- 2012 – 2019** **Studies and early industry experience.** B.Sc. Business Information Systems Engineering (2019) and B.Ed. English & French (2016), University of Augsburg. Backend development at Sigfox Germany GmbH (Munich, 2019); technical content and business development at Arrow Electronics (Frankfurt, 2017 – 2019).

Impact

- › *EsophagusVisualization* deployed at the University Hospital of Augsburg, adopted in active medical research
- › 6 first-author publications + 2 under review (full list at Google Scholar)
- › Best Position Paper Award, ICSOFT 2023
- › 5 supervised BSc and MSc theses across AI & healthcare
- › AI literacy across patient, legal, and public audiences

Languages

German native **English** full professional **French** professional